

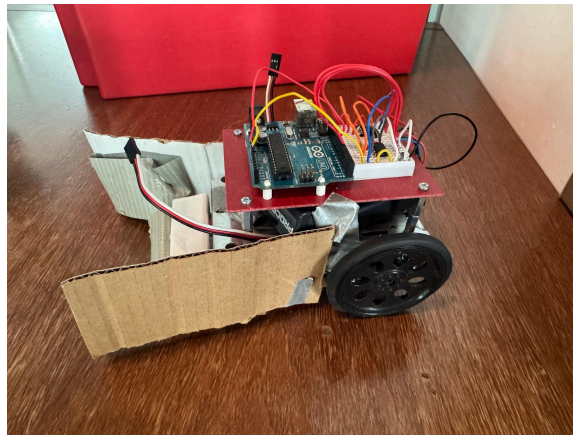
MAE 3780 Final Report Overview
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Robot Design and Strategy Overview

Our robots' mechanical design was a simple outward-facing "claw element" that utilized duct tape to capture the blocks and keep them within the unit. This was done using two cut-outs of cardboard that fit with the designated eight-inch parameters, with the inside lined with outward-facing duct tape. This design was meant to be able to flex outward due to the weight of the blocks once the competition started. The goal of our robot throughout this design was to keep it as simple as possible in order to avoid any technical difficulties on the day of the competition. This can be seen through the electrical design of our robot, which was only to utilize motor control through the H-bridge. We forwent using the color sensor or QTI's, as we felt this would overcomplicate the process and possibly lead to technical errors on the day of the competition. Finally, our software design also relied on simplicity in order to avoid technical errors or the possibility of blocks being pushed out of the boundary. Our final software design was to have a simple driving track that went straight until the color border, turned 115 degrees left, and then continued straight for a short period of time. This was meant to maximize the number of blocks that fit within the design parameters without having the possibility of losing blocks that we already captured

Design Process Reflection

Initially, our code was working on one of the days, but when we went back the next day, the robot wasn't functioning properly. We tried adjusting the color sensor closer to the ground, in case the sensor wasn't reading the color properly. We had to adjust the code slightly and ensure our sensor was close enough to the surface so that the light clearly illuminated the color panels, and then our robot followed the guidelines of actions it had to take. Our initial idea was to have a locomotive-esque snow plow, but that seemed to be unnecessary, so we scrapped that idea and just made a C-shape cardboard collector without any moving parts to simplify the process and design of the robot.



Design Process Reflection

Competition Analysis

On average, we collected around 1-5 cubes. During the round-robin stage, we managed to win a few rounds, tie some, and lose a few. The new set of batteries on the morning of the competition was dead, so we continued using the pack that was used for the milestones. During the competition, the structure and strength of the cardboard arms dramatically decreased. Each subsequent round had a decrease in performance, which led to the overall lower accumulation of cubes in the later rounds. The duct tape remained sticky for the entire duration of the competition. Our code was our biggest strength; we deployed simple code that ensured our robot stayed on the board without relying on the uncontrollable external factors, such as lighting. We found that other robots, as the lighting varied, would drive off the board, resulting in zero cubes collected.

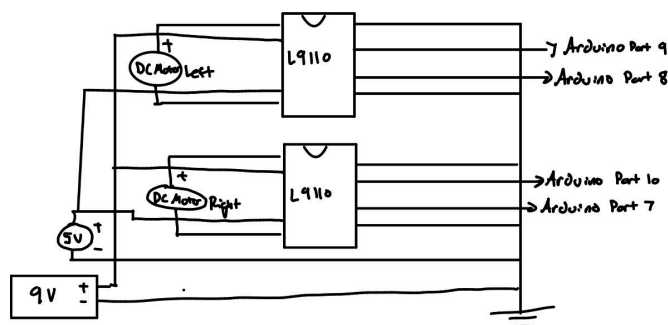
Conclusion

Our primary challenges were in programming and sensor calibration rather than the mechanical build. Performance issues during initial testing led us to lower the sensor's mounting height to improve color registration, which successfully resolved our navigation errors. We utilized a lightweight, C-shaped cardboard collector lined with tape, which proved effective in winning several matches. However, given the cardboard's flimsiness, a future iteration would replace it with a sturdier, 3D-printed sweeping mechanism that maintains a similar weight profile. For future students, prioritize being brilliant at the basics and focus on having a simple, yet effective design over an over-engineered piece of equipment that fails on game day.

Appendix A:

We did not purchase any outside materials to use within the project.

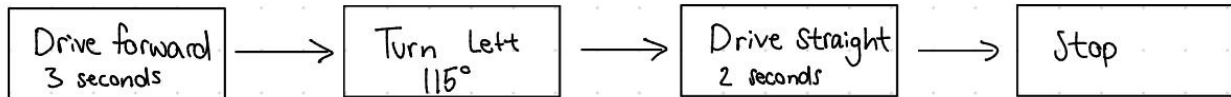
Appendix B:



Appendix C:

We did not utilize CAD within this assignment.

Appendix D:



Appendix E:

```
int main(void){ // setup code that only runs once
  DDRB = 0b00000111; // PB0, PB1, PB2 outputs
  DDRD = 0b10000000; // PD7 output

  while(1){ // code that loops forever

    PORTB = 0b00000110; // forward (PB1+PB2)
    PORTD = 0b00000000; // both directions forward

    _delay_ms(3000);

    PORTB = 0b00000110; // turn left
    PORTD = 0b00000000;
    PORTB |= 0b00000001; // left motor reverse (PB0)

    _delay_ms(2750);

    PORTB = 0b00000110; // forward
    PORTD = 0b00000000;

    _delay_ms(2000);

    PORTB = 0b00000000; //End Driving
    PORTD = 0b00000000;

    _delay_ms(100000);
  }
}
```